



Public Economics

Research Designs for Causal Inference

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Credit

This lecture is based on teaching material originally developed by **Alexander Ahammer** (Department of Economics, Johannes Kepler University Linz), from his course slides “*Forschungsdesigns: Daten und Forschungsdesigns in der Ökonomie*” (version 1 December 2025).

- Translated from German to English and lightly adapted for WU Vienna.
- Figures reproduced from external papers in the original slides are marked as placeholders here, pending the original image files.
- All illustrative diagrams and simulated-data examples are redrawn to match this course's visual style.

Experiments

Natural Experiments

Event Studies

Difference-in-Differences

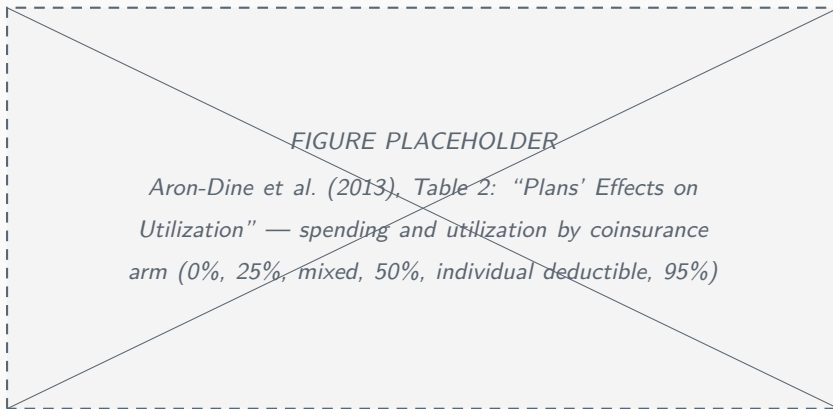
Instrumental Variables

Regression Discontinuity Design

Experiments

- We have seen that experiments (randomized control trials, RCTs) can solve the fundamental problem of causal inference.
- RCTs are the gold standard of causal inference.
- In the social sciences, RCTs are often difficult to implement:
 - A double-blind protocol is typically not possible
 - Too expensive
 - Conceptually possible but not practically feasible
 - Ethical concerns
 - Fairness
- Nonetheless, they are used increasingly often:
 - Social experiments for policy evaluation
 - Lab experiments in behavioral economics
 - Field experiments in development and labor economics
 - Increasingly common in other fields too

- How does cost-sharing (coinsurance/copayments) in health insurance affect health care spending?
- 1974–1981, ~5,800 randomly selected people (representative of the US population under 62) receive health insurance with a variable deductible/coinsurance rate.
- Cost: \$295 million; data collected mostly via surveys.
- Enormous influence on policy and the insurance industry.



Source: Aron-Dine, Einav & Finkelstein (2013), *Journal of Economic Perspectives* 27(1).

- Classic management problem: when monitoring is difficult or impossible, how can productivity best be maximized through incentive schemes?
- Does pay-for-performance (PFP) increase the productivity of existing employees (*incentive effect*) and attract more productive new employees (*selection effect*)?
 - PFP raises the marginal product of an extra unit of effort for the employee.
- Can we compare firms with different incentive schemes? No — these schemes are endogenous to firm success.
- We need experimental variation, ideally *within* a single firm.

- **Shearer (2004)**
- Tree planters in British Columbia (Canada), randomly assigned to fixed-wage vs. piece-rate pay on each workday.
- **Incentive effect:** 20% productivity gain on piece-rate days.
- But higher variance in productivity.
- No evidence of a selection effect.

- How important is neighborhood/environment for economic outcomes? A randomized mobility experiment.
- 4,600 low-income families with children, since 1994.
 - **Treatment:** housing vouchers for low-poverty areas, plus counseling.
 - **Control:** no vouchers.
- **Results:** better housing, few short-run effects — education unchanged, labor-market effects unclear.
- **Chetty et al. (2016):** matched with tax data, long-run effects on children — college attendance ↑, income ↑, single parenthood ↓.
- <http://www2.nber.org/mtopublic/>

- Social media has potential costs and benefits.
 - **Benefits:** lower cost of connecting, communicating, and sharing information.
 - **Costs:** negative correlations with mental health, cyberbullying, political polarization.
- **Allcott et al. (2020, AER):** RCT with 2,743 Facebook users, before the 2018 midterm elections; the treatment group was paid to deactivate their account.
- **Results:**
 - Less online activity, more TV watching and social activities
 - Less “news knowledge” and political polarization
 - Higher subjective well-being
 - Long-lasting reduction in Facebook use

- With experiments, the workload is highest *before* the first results are available.
- Analysis of experimental data is relatively simple.
- If randomization worked, comparing means between the treatment and control group is sufficient.
- OLS regressions can control for additional variables:

$$Y_i = \beta_0 + \beta_1 D_i + \gamma X + \varepsilon_i \quad (1)$$

- **Non-random assignment**
 - Test baseline characteristics between treatment (T) and control (C) groups
- **Non-compliance**
 - C also gets treated
 - T does not take up the treatment
- **Attrition**
 - Participants leave the experiment
- **Hawthorne effect / experimenter demand effects**
 - Participants behave differently because they know they are part of an experiment

Natural Experiments

- We are often interested in questions that cannot be answered through experiments.
- In (non-experimental) observational data, this is possible — but we need a **design**.
- We primarily look for so-called **natural experiments**.
 - Synonyms: quasi-experiments, accidental experiments
- Natural experiments are circumstances that cause otherwise identical people to receive a treatment quasi-randomly, while others do not:
 - Nature (e.g. weather, multiple births, health shocks)
 - (Arbitrary) administrative rules (e.g. Maimonides' rule, age cutoffs)
 - (Unpredictable) changes in law (e.g. minimum wages)
 - Geographic variation/boundaries (e.g. broadband internet access)
 - Natural disasters

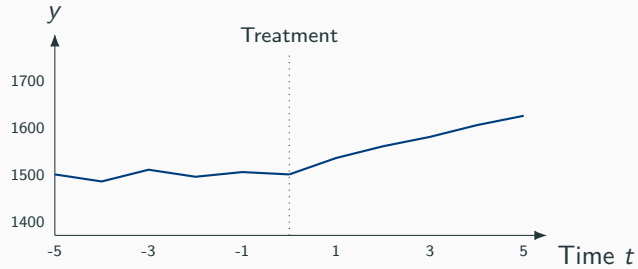
- To analyze natural experiments, we need additional econometric methods:
 - Event studies
 - Difference-in-differences
 - Instrumental variables
 - Regression discontinuity designs

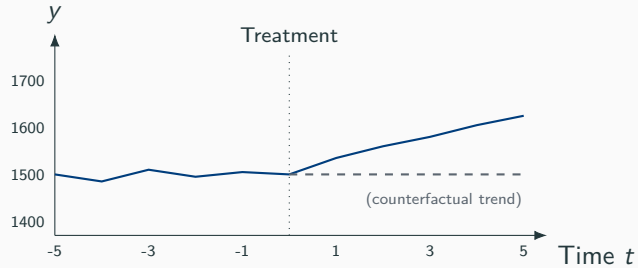
Trade-off

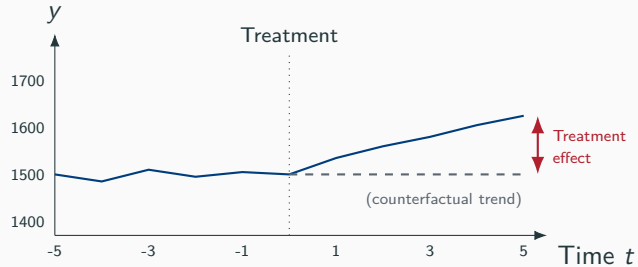
Pros: cheap to carry out. **Cons:** hard to find, external validity concerns.

Event Studies

- **Setting:** we have an outcome y that varies over time.
 - Time-series or panel data required.
- Treatment occurs at time $t = 0$ (e.g. a policy change).
- **Idea:** everything that happens before $t = 0$ is unrelated to the treatment. Everything that happens after $t = 0$ is conditioned on the treatment.
- If the outcome trend is “flat” before $t = 0$, we can make an assumption about the counterfactual:
 - How would the trend have evolved *without* the treatment?







- The counterfactual trend is an **assumption**.
 - Not testable → the fundamental problem of causal inference.
- If we are willing to assume that, absent treatment, the trend would have simply continued, the difference between the blue and gray lines is the treatment effect.
- Another important assumption: nothing *else* may happen at $t = 0$ that could affect the outcome.
 - Must be argued for.
 - Requires institutional knowledge.

Estimation with a simple OLS regression:

$$y_{it} = \beta_0 + \delta \text{post}_{it} + \varepsilon_{it} \quad (2)$$

- post_{it} is a dummy variable equal to 1 in every period $t > 0$ and 0 in prior periods.
- δ is the treatment effect (the difference in y -means between the pre- and post-treatment period).

If we are interested in **dynamic effects**, estimate with period dummies:

$$\begin{aligned} y_{it} &= \beta_0 + \delta_{-5}t_{i,-5} + \delta_{-4}t_{i,-4} + \cdots + \delta_{-1}t_{i,-1} + \delta_1t_{i,1} + \delta_2t_{i,2} + \cdots + \delta_5t_{i,5} + \varepsilon_{it} \\ &= \beta_0 + \sum_{k=-5 \mid k \neq 0}^5 \delta_k t_{ik} + \varepsilon_{it} \end{aligned} \quad (3)$$

- We define t_0 as the base period and therefore omit its dummy.
- The coefficients $\hat{\delta}_k$ measure the difference between y_k in each period k and y_0 at $t = 0$.

```
. list id t post y in 1/22, sepby(id)
```

	id	t	post	y
1.	1	-5	0	1500.392
2.	1	-4	0	1458.272
3.	1	-3	0	1668.017
4.	1	-2	0	1546.392
5.	1	-1	0	1425.803
6.	1	0	0	1533.735
7.	1	1	1	1492.77
8.	1	2	1	1618.688
9.	1	3	1	1635.518
10.	1	4	1	1582.649
11.	1	5	1	1682.446

```
[... id 2 omitted here, same structure ...]
```

```
. ttest y, by(post)
```

Two-sample t test with equal variances

Group	Obs	Mean	Std. Err.	Std. Dev.	[95% Conf. Interval]	
0	6,000	1498.789	1.04265	80.76335	1496.746	1500.833
1	5,000	1576.126	1.227076	86.76741	1573.721	1578.532
combined	11,000	1533.943	.8770979	91.99081	1532.223	1535.662
diff		-77.33677	1.599783		-80.47264	-74.20091

diff = mean(0) - mean(1)

t = -48.3421

Ho: diff = 0

degrees of freedom = 10998

Ha: diff < 0

Ha: diff != 0

Ha: diff > 0

Pr(T < t) = 0.0000

Pr(|T| > |t|) = 0.0000

Pr(T > t) = 1.0000

```
. reg y post
```

Source	SS	df	MS	Number of obs	=	11,000
				F(1, 10998)	=	2336.95
Model	16311754.6	1	16311754.6	Prob > F	=	0.0000
Residual	76765177.5	10,998	6979.92158	R-squared	=	0.1753
				Adj R-squared	=	0.1752
Total	93076932.1	10,999	8462.30858	Root MSE	=	83.546

y	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
post	77.33677	1.599783	48.34	0.000	74.20091	80.47264
_cons	1498.789	1.078573	1389.60	0.000	1496.675	1500.904

⇒ Without control variables, the OLS estimator equals the difference in means between the pre- and post-treatment period.

⇒ The estimated treatment effect is **77.3**.

```
. quietly tabulate t, gen(dt_)  
. list id t dt_* in 1/22, sepby(id)
```

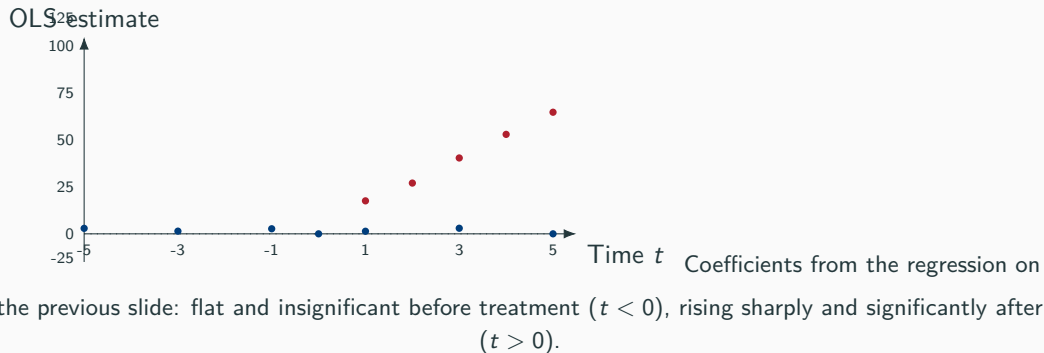
	id	t	dt_1	dt_2	dt_3	dt_4	dt_5	dt_6	dt_7	dt_8	dt_9	dt_10	dt_11
1.	1	-5	1	0	0	0	0	0	0	0	0	0	0
2.	1	-4	0	1	0	0	0	0	0	0	0	0	0
3.	1	-3	0	0	1	0	0	0	0	0	0	0	0
4.	1	-2	0	0	0	1	0	0	0	0	0	0	0
5.	1	-1	0	0	0	0	1	0	0	0	0	0	0
6.	1	0	0	0	0	0	0	1	0	0	0	0	0
7.	1	1	0	0	0	0	0	0	1	0	0	0	0
8.	1	2	0	0	0	0	0	0	0	1	0	0	0
9.	1	3	0	0	0	0	0	0	0	0	1	0	0
10.	1	4	0	0	0	0	0	0	0	0	0	1	0
11.	1	5	0	0	0	0	0	0	0	0	0	0	1

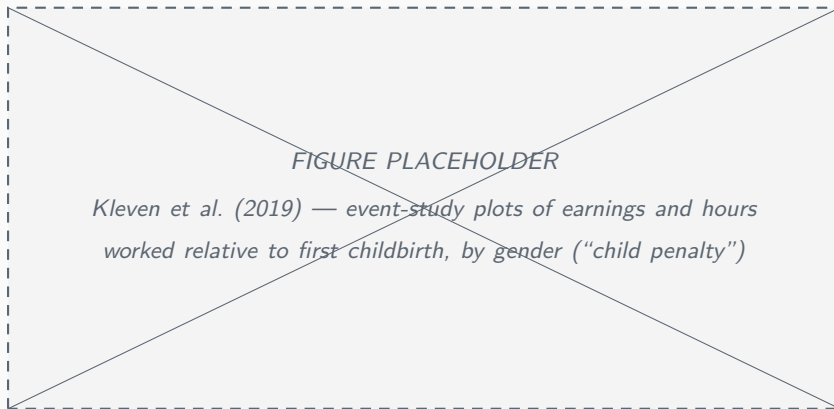
Simulated Example: Equation (3) by OLS

```
. reg y dt_1-dt_5 dt_7-dt_11
```

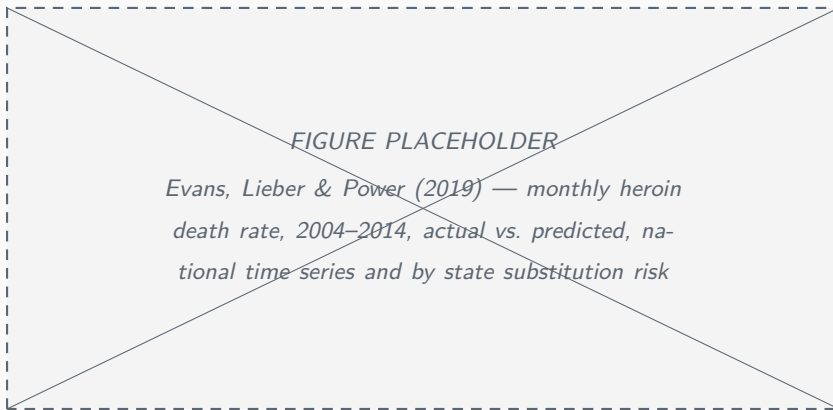
Source	SS	df	MS	Number of obs	=	11,000
				F(10, 10989)	=	343.21
Model	22151716.7	10	2215171.67	Prob > F	=	0.0000
Residual	70925215.4	10,989	6454.20105	R-squared	=	0.2380
				Adj R-squared	=	0.2373
Total	93076932.1	10,999	8462.30858	Root MSE	=	80.338

y	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
dt_1	5.825865	3.592826	1.62	0.105	-1.216721	12.86845
dt_2	2.872052	3.592826	0.80	0.424	-4.170535	9.914638
dt_3	5.407958	3.592826	1.51	0.132	-1.634629	12.45054
dt_4	2.766989	3.592826	0.77	0.441	-4.275597	9.809575
dt_5	5.954441	3.592826	1.66	0.097	-1.088145	12.99703
dt_7	35.16286	3.592826	9.79	0.000	28.12027	42.20544
dt_8	54.08144	3.592826	15.05	0.000	47.03886	61.12403
dt_9	80.86031	3.592826	22.51	0.000	73.81772	87.90289
dt_10	106.0099	3.592826	29.51	0.000	98.96728	113.0524
dt_11	129.5922	3.592826	36.07	0.000	122.5496	136.6347
_cons	1494.985	2.540512	588.46	0.000	1490.005	1499.965





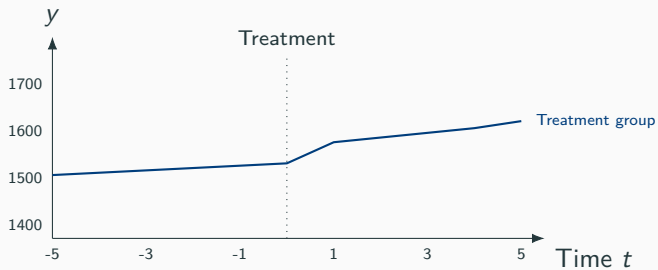
Source: Kleven, Landais & Sørensen (2019), *AEJ: Applied Economics* 11(4).

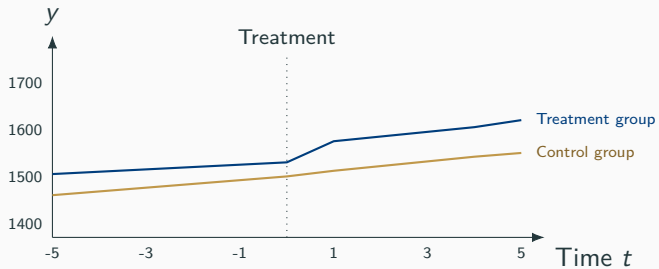


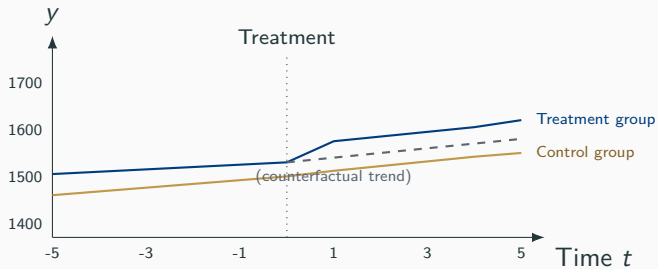
Source: Evans, Lieber & Power (2019), "How the Reformulation of OxyContin Ignited the Heroin Epidemic," *Review of Economics and Statistics* 101(1).

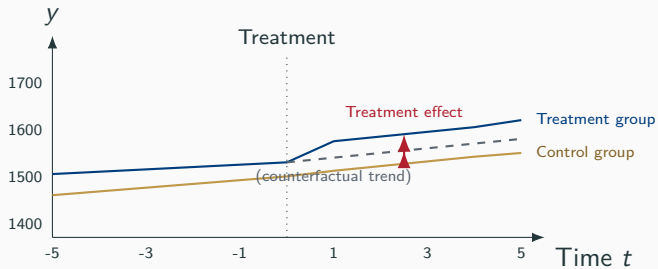
Difference-in-Differences

- The assumptions behind event studies are often not satisfied, because we cannot rule out that multiple treatments occur at the same time (t_0).
- A **control group** can help here.
 - A group that did not receive the treatment.
 - E.g. because of administrative rules, legal changes applying to only part of the population, etc.
- **Difference-in-differences (DD)** estimation:
 - 1st difference: before/after treatment (as in event studies)
 - 2nd difference: between treatment and control group
- Only possible if the treatment and control group move on **parallel trends** before the treatment.

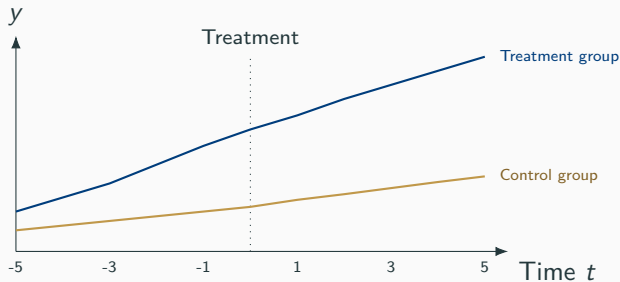








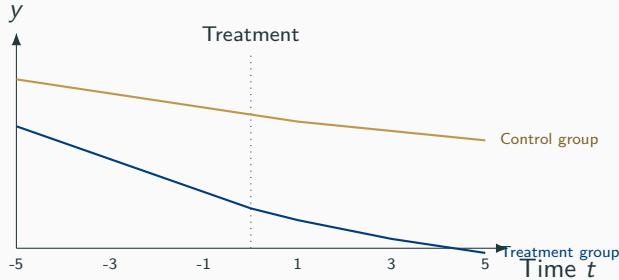
- DD assumes that the control group's path *is* the counterfactual trend for the treatment group.
- **Common trends assumption:** absent treatment, the treatment group would have moved on the *same* path as the control group.
 - Not testable, but plausible if trends are parallel *before* the treatment.



⇒ A DD estimate would **not** be credible

here, because the treatment group is already rising much faster than the control group *before* the treatment (no parallel trends).

When DD Does *Not* Work: Non-Parallel (Crossing) Trends



⇒ A DD estimate would also **not** be possible here, because the pre-treatment trends are not parallel (they cross).

Under the common trends assumption, the treatment effect can be estimated with the following OLS regression:

$$y_{it} = \beta_0 + \beta_1 \textit{treated}_i + \beta_2 \textit{post}_{it} + \delta (\textit{treated}_i \times \textit{post}_{it}) + \varepsilon_{it} \quad (4)$$

- $\textit{treated}_i$ is a dummy indicating whether an observation is in the treatment ($\textit{treated}_i = 1$) or control group ($\textit{treated}_i = 0$).
- In the simplest case, $\hat{\delta}$ is the estimated difference in post-treatment means between the treatment and control group, net of the time-constant difference in y between the two groups.

And to get dynamic effects:

$$y_{it} = \beta_0 + \beta_1 \text{treated}_i + \sum_{j=-5 | j \neq 0}^5 \tau_j t_{ij} + \sum_{k=-5 | k \neq 0}^5 \delta_k (\text{treated}_i \times t_{ik}) + \varepsilon_{it} \quad (5)$$

- The coefficients $\hat{\delta}_k$ measure the difference between the treatment and control group in each period k , relative to their difference in period 0.

```
. list id t post treated y in 1/22, sepby(id)
```

	id	t	post	treated	y
1.	1	-5	0	0	1463.392
2.	1	-4	0	0	1433.272
3.	1	-3	0	0	1655.017
4.	1	-2	0	0	1545.392
5.	1	-1	0	0	1436.803
6.	1	0	0	0	1556.735
7.	1	1	1	0	1501.72
8.	1	2	1	0	1612.518
9.	1	3	1	0	1609.639
10.	1	4	1	0	1564.446
11.	1	5	1	0	1638.188

```
[... id 2 (treated = 1) omitted here, same structure ...]
```

```
. generate treatedXpost = treated*post
```

```
. reg y treated post treatedXpost
```

Source	SS	df	MS	Number of obs	=	11,000
				F(3, 10996)	=	1747.29
Model	35405865.2	3	11801955.1	Prob > F	=	0.0000
Residual	74271885.9	10,996	6754.44578	R-squared	=	0.3228
				Adj R-squared	=	0.3226
Total	109677751	10,999	9971.61115	Root MSE	=	82.185

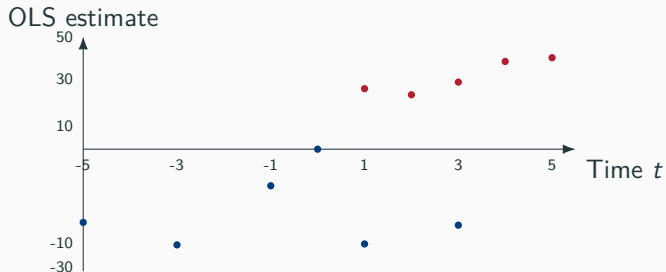
y	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
treated	56.17291	2.122019	26.47	0.000	52.01337	60.33245
post	73.14872	2.225592	32.87	0.000	68.78616	77.51128
treatedXpost	30.436	3.147463	9.67	0.000	24.26641	36.60559
_cons	1488.703	1.500494	992.14	0.000	1485.762	1491.644

⇒ How should the constant and the coefficients on treated and post be interpreted?

⇒ The estimated treatment effect is **30.4**.

```
. g t100 = t+100
. reg y b100.t100##i.treated
    Number of obs = 11,000    F(21, 10978) = 288.32    R-squared = 0.3555
```

	y	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]
t100						
95	-51.37909	5.075141	-10.12	0.000	-61.32729	-41.4309
99	-2.968469	5.075141	-0.58	0.559	-12.91666	6.979722
[...]						
101	30.71617	5.075141	6.05	0.000	20.76798	40.66437
105	70.0025	5.075141	13.79	0.000	60.05431	79.95069
1.treated	61.19874	5.075141	12.06	0.000	51.25054	71.14693
t100#treated						
95 1	-5.590081	7.177334	-0.78	0.436	-19.65895	8.478786
99 1	-6.15418	7.177334	-0.86	0.391	-20.22305	7.914687
[...]						
101 1	22.86918	7.177334	3.19	0.001	8.800316	36.93805
105 1	29.53941	7.177334	4.12	0.000	15.47054	43.60827
_cons	1512.386	3.588667	421.43	0.000	1505.351	1519.42



Coefficients on $treated_i \times t_{ik}$ from the regression above: flat and insignificant before treatment, positive and significant after.

- **Ahammer, Pruckner & Stifter (202x)**
- Breast cancer $\implies$ health care spending, employment, wages
- Breast cancer = shock; women without breast cancer serve as the control group
 - The control group receives a placebo shock based on the age-pdf
- Model:

$$y_{it} = \beta_0 + \beta_1 \text{breastcancer}_i + \sum_{j=-20 | j \neq -1}^{20} \tau_j t_{ij} + \sum_{k=-20 | k \neq -1}^{20} \delta_k (\text{breastcancer}_i \times t_{ik}) + \varepsilon_{it} \quad (6)$$

FIGURE PLACEHOLDER

*Ahammer, Pruckner & Stiftinger: level and change in
healthcare expenditures, quarters relative to first (placebo)
diagnosis, breast cancer patients vs. other women*

FIGURE PLACEHOLDER

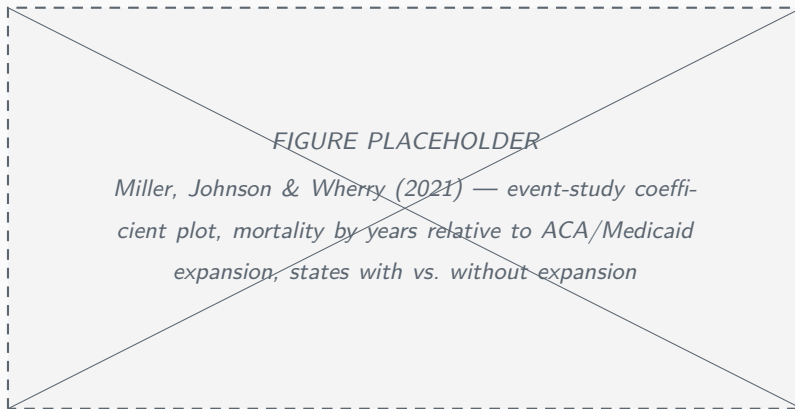
*Ahammer, Pruckner & Stifter: level and change
in days employed, quarters relative to first (placebo)
diagnosis, breast cancer patients vs. other women*

FIGURE PLACEHOLDER

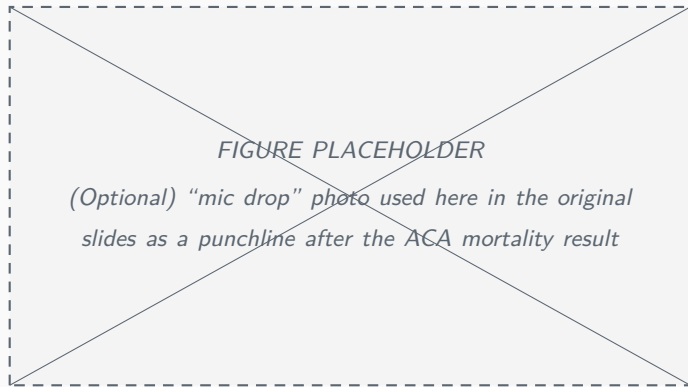
*Ahammer, Pruckner & Stifter: annual wage
conditional on working and change in annual
wage, quarters relative to first (placebo) diagnosis*

- **Miller, Johnson & Wherry (2021)**
- Affordable Care Act / Medicaid expansion $\implies$ mortality?
- (Lack of) health insurance is a major issue in the US — did Obamacare (= ACA) help?
- Expansion did not happen simultaneously in all states; states that had not yet expanded serve as the control group.
- Model:

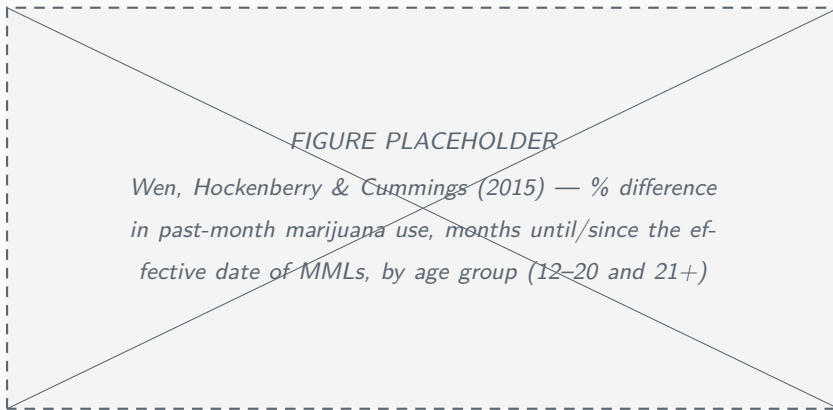
$$mortality_{it} = \beta_0 + \beta_1 statetreated_i + \sum_{j=-6 | j \neq 0}^3 \tau_j t_{ij} + \sum_{k=-6 | k \neq 0}^3 \delta_k (statetreated_i \times t_{ik}) + \varepsilon_{it} \quad (7)$$



Source: Miller, Johnson & Wherry (2021), *Review of Economics and Statistics* 103(3).



- **Wen, Hockenberry & Cummings (2015)**
- Medical marijuana laws (MMLs); the gradual legalization of marijuana use for medical purposes in the US.
- How does legalization affect marijuana use among young people?
- States without MMLs serve as the control group.



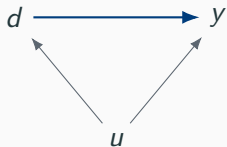
Source: Wen, Hockenberry & Cummings (2015), *NBER Working Paper* 20248.

Instrumental Variables

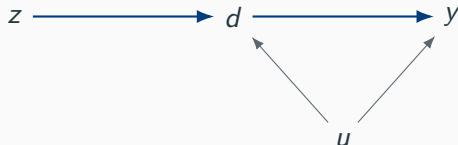
We are interested in the following model:

$$y = \beta_0 + \beta_1 d + \varepsilon \quad (8)$$

- But we know the relationship is not causal, e.g. because of unobservable third variables or reverse causality.
- In an RCT, we would randomize d and thereby estimate a causal effect.
- **Idea of an instrumental variable:** we look for a variable that performs this randomization for us.



An unobserved variable u confounds both d and y — the OLS estimate of $d \rightarrow y$ is biased.



- z is an **instrumental variable**.
- We can identify the effect of d on y caused by z , without controlling for u .
- **Most important assumption:** there is no direct effect $z \rightarrow y$ that does not run through d (**exclusion restriction**).

- **Angrist & Krueger (1991)**
- Returns to schooling: a causal effect is hard to estimate because the schooling decision is endogenous.
- Compulsory schooling duration $\rightarrow$ schooling decision $\rightarrow$ income.
- Changes in compulsory schooling duration: birth date (quasi-random!) determines whether one can leave school at 15 or 16.
- The design estimates the effect of additional schooling induced by an increase in compulsory schooling duration (external validity).

- **Aizer & Doyle Jr. (2015)**
- How does incarceration affect recidivism?
- Judge strictness $\rightarrow$ incarceration $\rightarrow$ probability of reoffending.
- Random assignment to judges in the US — some are stricter than others.
- The design estimates the effect of *marginal* incarceration: what is the recidivism rate among people who end up incarcerated only because they happened to draw a strict judge?

Estimation using a two-stage model (two-stage least squares, 2SLS):

1. **First stage:**

$$d_i = \alpha_0 + \alpha_1 z_i + \xi_i \quad (9)$$

2. Compute the fitted value $\hat{d}_i$ for all i .

3. **Second stage:**

$$y_i = \beta_0 + \beta_1 \hat{d}_i + \varepsilon_i \quad (10)$$

- Standard errors in the second stage must be adjusted, because $\hat{d}_i$ is itself estimated from the data.
- Identifies the so-called **local average treatment effect (LATE)**.

```
. clear
. set obs 10000
number of observations (_N) was 0, now 10,000
. g z = rnormal(50, 10)
. g d = 5 + .5*z + rnormal(0, 1.5)
. g y = 2.5 + .25*d + rnormal(0, 2)
. reg d z
```

Source	SS	df	MS	Number of obs	=	10,000
				F(1, 9998)	>	99999.00
Model	248031.051	1	248031.051	Prob > F	=	0.0000
Residual	22959.4347	9,998	2.29640275	R-squared	=	0.9153
				Adj R-squared	=	0.9153
Total	270990.486	9,999	27.1017588	Root MSE	=	1.5154

	d	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]
	z	.5009801	.0015244	328.65	0.000	.497992 .5039681
	_cons	4.954964	.0777234	63.75	0.000	4.80261 5.107317

```
. predict d_hat, xb
```

```
. reg y d_hat
```

Source	SS	df	MS	Number of obs	=	10,000
				F(1, 9998)	=	3665.97
Model	15115.1207	1	15115.1207	Prob > F	=	0.0000
Residual	41222.6029	9,998	4.12308491	R-squared	=	0.2683
				Adj R-squared	=	0.2682
Total	56337.7236	9,999	5.6343358	Root MSE	=	2.0305

y	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
d_hat	.2468612	.0040772	60.55	0.000	.2388691	.2548532
_cons	2.572058	.1240221	20.74	0.000	2.328949	2.815166

Note: the point estimate is correct, but the standard errors here are *not* — they need the 2SLS correction on the next slide.

```
. ivreg2 y (d=z)
IV (2SLS) estimation                Number of obs = 10000    F(1,9998) = 3801.35

            y            Coef.      Std. Err.      z    P>|z|      [95% Conf. Interval]
      d          .2468612    .0040035     61.66   0.000     .2390144     .2547079
    _cons        2.572058    .1217816     21.12   0.000     2.33337     2.810745

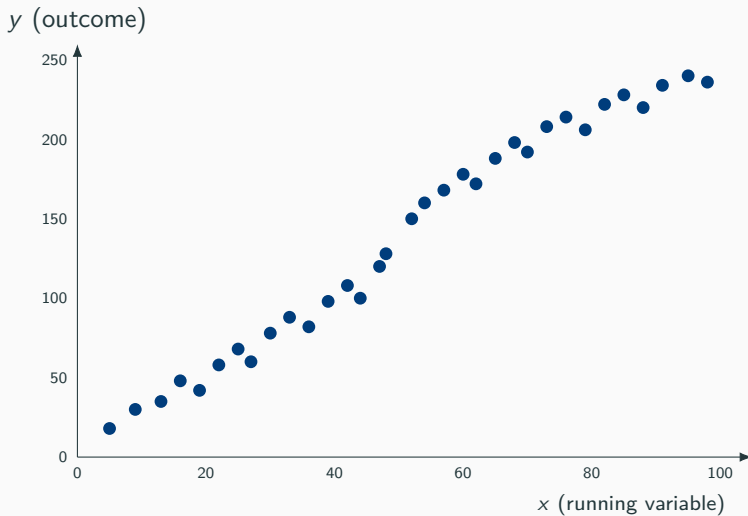
Underidentification test (Anderson canon. corr. LM statistic):          9152.759
                                Chi-sq(1) P-val =                      0.0000

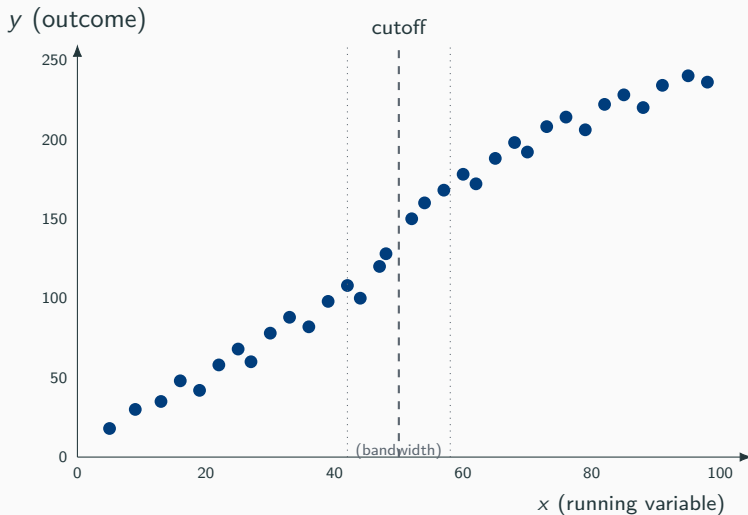
Weak identification test (Cragg-Donald Wald F statistic):              1.1e+05
Stock-Yogo weak ID test critical values: 10% maximal IV size          16.38

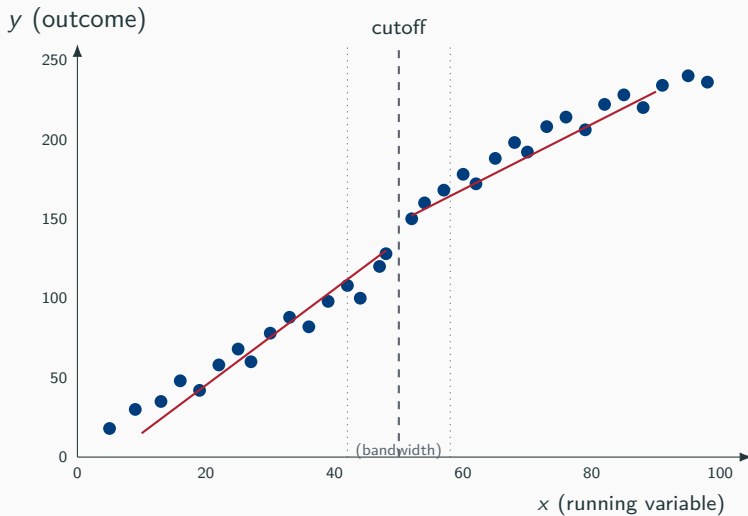
Instrumented:      d
Excluded instruments: z
```

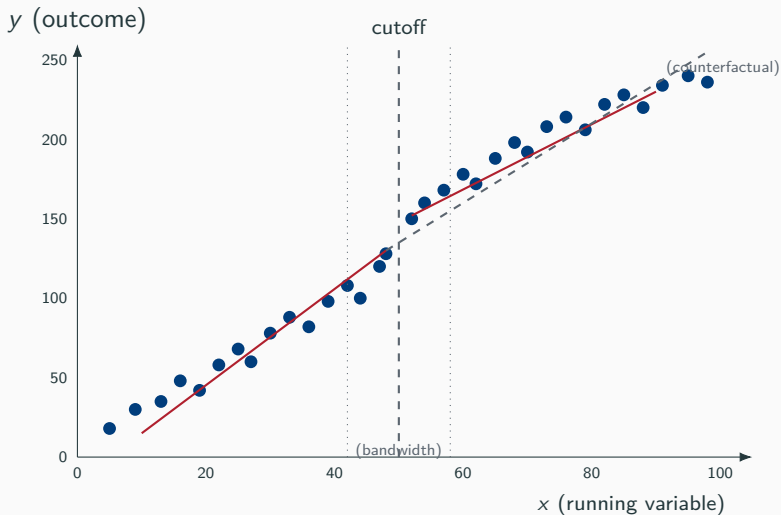
Regression Discontinuity Design

- **Setting:** treatment is assigned via a cutoff.
 - Unemployed workers over 40 receive unemployment benefits for longer.
 - Students with an ACT score above 35 are admitted to Harvard.
 - Newborns weighing less than 2,500g receive special treatment.
- **Idea:** observations in a very small window around the cutoff (= **bandwidth**) are as good as randomly assigned.
 - You cannot choose whether you become unemployed at age 39.9 or 41.1.
 - It is quasi-random whether you score 34.9 or 35 on the ACT.
 - It is quasi-random whether birth weight is 2,499g or 2,500g.
- **Most important assumption:** you cannot choose whether you fall above or below the cutoff (**no manipulation assumption**).
- A *local* experiment.









Estimation is possible with OLS:

$$y_i = \beta_0 + \beta_1(x_i - c_0) + \delta d_i + \varepsilon_i \quad (11)$$

where c_0 is the cutoff (50 in the previous example) and $d_i = \mathbf{1}\{x_i > c_0\}$.

Modeling decisions:

- **Bandwidth** (the smaller, the “better” the local experiment — but the higher the data requirement)
- **Bin size** (the x -axis is typically “binned”)
- **Trends** (the running variable is modeled linearly here; a higher-order polynomial may be necessary)

Validity tests: are covariates “smooth” around the cutoff? Is there “bunching” at or below the cutoff?

- **Nekoei & Weber (2009)**
- Unemployment benefits are extended by 9 weeks if the worker is ≥ 40 years old and worked in 6 of the last 10 years.
- **Mechanism:** signalling vs. more time to search for better jobs.
- Effects on labor-market outcomes.
- **Extension:** effects on health (Ahammer & Packham 2023).

FIGURE PLACEHOLDER

*Nekoei & Weber (2009), Panel A — density of age at layoff
around the age-40 cutoff (smoothness/no-manipulation check)*

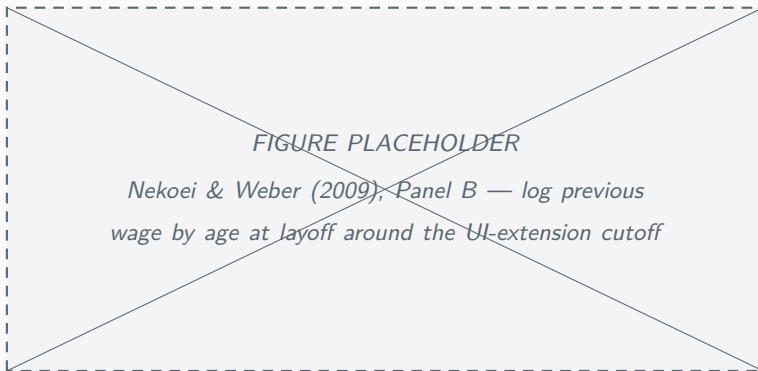
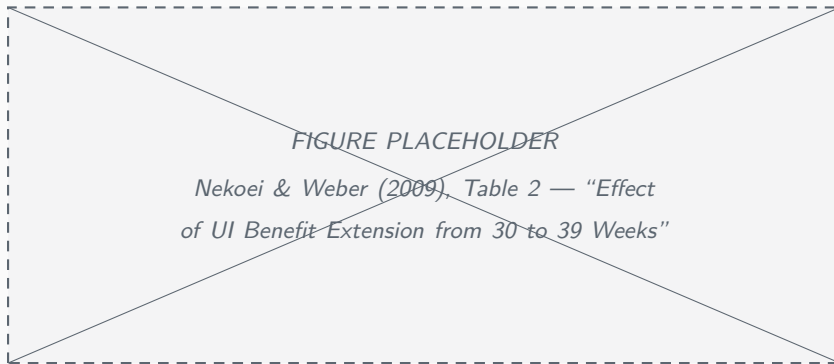
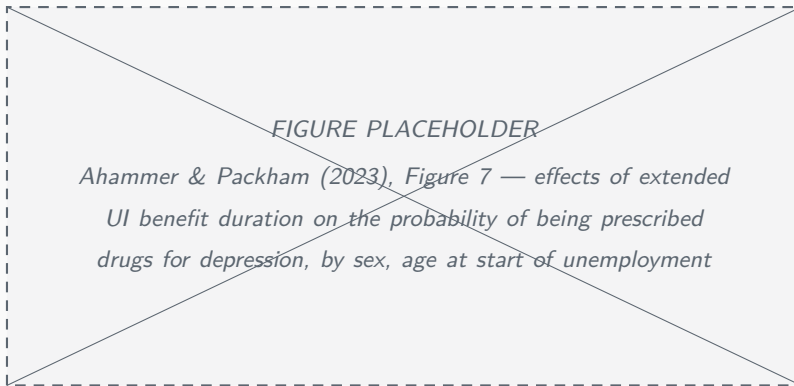


FIGURE PLACEHOLDER

Nekoei & Weber (2009) — four-panel RDD results: nonemployment duration, probability of finding a job within 39 weeks, wage change between jobs, and new wage $> 0.5 \times$ old wage, all vs. age at layoff



Source: Nekoei & Weber (2009/2017), *American Economic Review* 107(2).



Source: Ahammer & Packham (2023).

Any questions?

